SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA0612 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:

I K. Pujitha (192524149) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Pujitha

Annexure A

Short Answer Test

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 Marks)

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors. (5 Marks)

4.A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms. (5 Marks)

5 .During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

nple Perceptron Model

Given,

- Study hours (x_1) = 2
- Attendance (x_2) = 1
- weight (w_1) = 0.6
- Bias (b) = -0.5

Step1 : Calculate the net input

Formula:

$$Net = (x_1 w_1) + (x_2 w_2) + b$$

Substituting the values:

$$Net = (2 \times 0.6) + (1 \times 0.4) - 0.5$$

$$Net = 1.2 + 0.4 - 0.5$$

$$Net = 1.1$$

Step2 : Apply the step activation function

The step function is :

$$f(Net) = \begin{cases} 1, & Net \geq 0 \\ 0, & Net < 0 \end{cases}$$

Since $1.1 \geq 0$, the Output is 1.

Conclusion: The student Passes, because the Perceptron Output is 1.

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3 weight update using the Delta Rule

Given,

- Initial weight (w) = 0.5
- Learning rate (η) = 0.1

- Input (x) = 2
- Actual Output (O) = 0.6
- Target output (T) = 1

step1: Calculate the Output Error

formula:

$$\text{Error} = T - O$$

$$\text{Error} = 1 - 0.6$$

$$= 0.4$$

step2: Calculate the weight change using the delta rule:

$$\Delta w = \eta (T - O) x$$

Substituting:

$$\Delta w = 0.1 (1 - 0.6)(2)$$

$$\Delta w = 0.1 \times 0.4 \times 2$$

$$\Delta w = 0.08$$

step3: Calculate the new weight

formula:

$$w_{new} = w_{old} + \Delta w$$

$$w_{new} = 0.5 + 0.08$$

$$w_{new} = 0.58$$

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3) Evaluation of using a confusion matrix:

Given.

- True Positives (TP) = 180
- True Negatives (TN) = 150
- False Positives (FP) = 20

False Negatives (FN) = 50

Step 1: Calculate accuracy

Accuracy measures the overall correctness of the model.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

$$= \frac{180 + 150}{180 + 150 + 20 + 50} \times 100$$

$$= \frac{330}{400} \times 100$$

Accuracy = 82.5%

Step 2: Calculate Precision

Precision measures how many Predicted Positive cases are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP} \times 100$$

$$= \frac{180}{180 + 20} \times 100$$

Precision = 90%

Step 3: Calculate sensitivity (Recall)

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100$$

$$= \frac{180}{180 + 50} \times 100$$

Sensitivity = 78.26%.

step 4:- Calculate specificity

$$\text{Specificity} = \frac{TN}{TN+FP} \times 100$$

$$= \frac{150}{150+20} \times 100$$

Specificity = 88.24%.

Conclusion: The model has Good precision and specificity, but its sensitivity is relatively low.

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4. Genetic Algorithm for Maximizing $f(x) = x^2$

Given initial population 2, 4, 6, 8

step 1:- Calculate fitness values

The fitness function is

$$f(x) = x^2$$

Individual (x)	Fitness (x^2)
2	4
4	16
6	36
8	64

Conclusion:- Individual 8 has highest fitness value of 64, so it is the best candidate in the initial Population.

Step 2: selection

Individuals with higher fitness are more likely to be selected as Parents.

Step 3: Crossover

Convert the selected Parents into 4-bit binary Representations:

- 6 = 0110
- 8 = 1000

Parent 1 : 0 | 110

Parent 2 : 1 | 000

Child 1 : 0 | 000 = 0

Child 2 : 1 | 110 = 14

The offspring are 0 and 14.

The fitness of child 2 is

$$f(14) = 14^2 = 196$$

Step 4: Role of mutation.

mutation randomly changes one or more bits is an offspring.

Importance :

1) maintains diversity in the Population.
2) Prevents the algorithm from Getting stuck at a local optimum.
3) Introduces new solutions.

5) Training Accuracy and validation Accuracy

Given,

- Training accuracy increases from 60% to 95%.
- validation accuracy decreases from 72% to 68%.

Step1: Analyze the trend

Initially, both training and validation accuracy increase, showing that the model is learning useful Patterns.

After a certain number of Epochs:

- Training accuracy continues increasing to 95%.
- validation accuracy decreases from 72% to 68%.

Step2: Identify the problem

The problem is overfitting.

overfitting occurs when a neural network learns the training data too closely, including noise and unnecessary details, fails to Generalize well to new data.

Step3: Importance of validation accuracy

validation accuracy helps us:

1) measure how well the model performs on unseen data.
2) Detect overfitting
3) select the best model and number of Epochs.
4) Compare different model settings.